EXPERIMENT-9

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USER INTERFACE AND DESIGN

Form Validation using HTML, CSS, JavaScript (Validator.js)

AIM

To design input forms that validate user data such as email and phone number, and display appropriate error messages using HTML, CSS, JavaScript, and Validator.js.

TOOLS USED

- HTML
- CSS
- JavaScript
- Validator.js (CDN)

STEP 1: Setting Up the HTML Form

```
<!DOCTYPE html>
<html>
<head>
<title>Form Validation</title>
</head>
<body>

<h2>Signup Form</h2>

<form id="form">
<label>Email:</label><br>
<input type="text" id="email"><br>
```

```
<small id="emailError"></small><br><br>

<label>Phone Number:</label><br>
<input type="text" id="phone"><br>
<small id="phoneError"></small><br><br>

<button type="submit">Submit</button>
</form>
</body>
</html>
```

Output screen:

Explanation

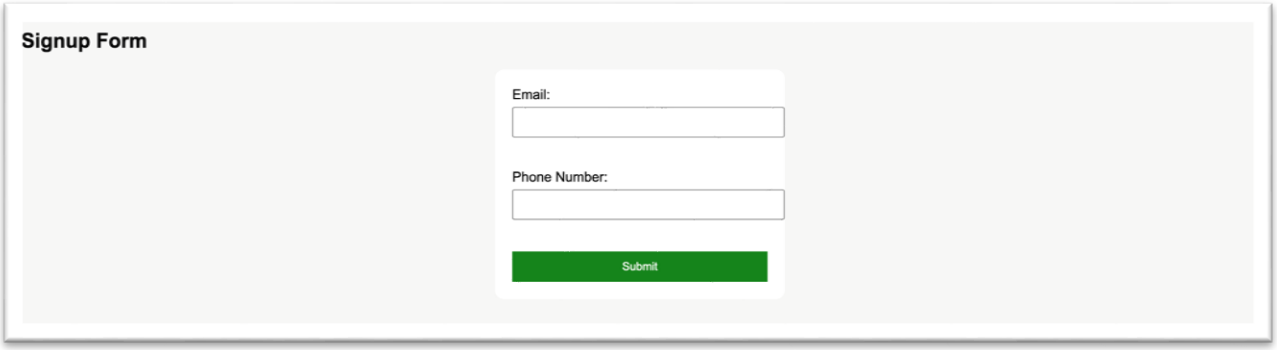
- A basic HTML form is created with:
  - Email input field
  - Phone number input field
  - Submit button
- <small> tags are used to display error messages
- No validation or styling yet

STEP 2: Styling the Form with CSS

```
<!DOCTYPE html>
<html>
<head>
<title>Form Validation</title>
<style>
body {
font-family: Arial;
background-color: #f4f4f4;
```

```
}

form {
width: 300px;
margin: auto;
padding: 20px;
background: white;
border-radius: 10px;
}

input {
width: 100%;
padding: 8px;
margin-top: 5px;
}

button {
width: 100%;
padding: 10px;
background: green;
color: white;
border: none;
}

small {
color: red;
}
</style>
</head>
<body>

<h2>Signup Form</h2>

<form id="form">
<label>Email:</label><br>
<input type="text" id="email"><br>
<small id="emailError"></small><br><br>

<label>Phone Number:</label><br>
<input type="text" id="phone"><br>
<small id="phoneError"></small><br><br>

<button type="submit">Submit</button>
</form>
</body>
</html>
```

Output Screen:



Explanation

- CSS improves visual appearance:
  - Centered form layout
  - Background color added
  - Styled inputs and button
- Error messages are shown in **red color**
- Still no validation logic applied

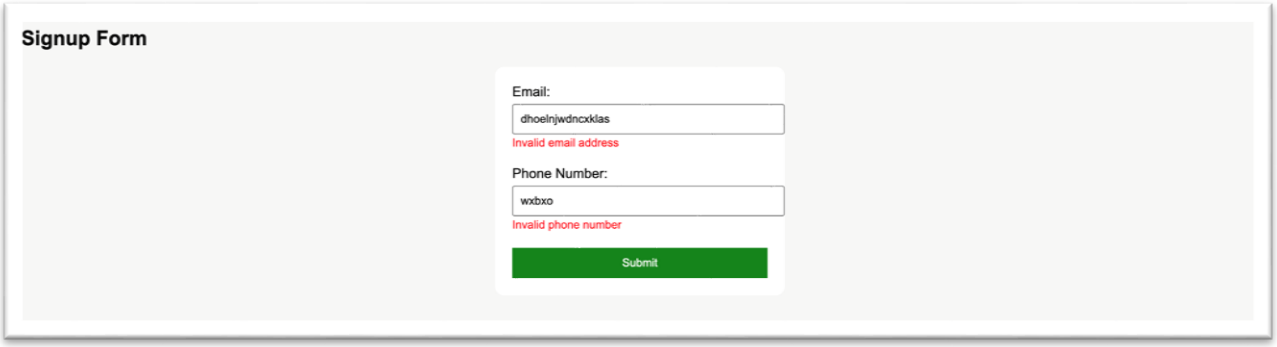
STEP 3: JavaScript Validation using Validator.js

```
<!DOCTYPE html>
<html>
<head>
<title>Form Validation</title>
<style>
body {
font-family: Arial;
background-color: #f4f4f4;
}

form {
width: 300px;
margin: auto;
padding: 20px;
background: white;
border-radius: 10px;
}

input {
width: 100%;
padding: 8px;
margin-top: 5px;
```

```
}

button {
width: 100%;
padding: 10px;
background: green;
color: white;
border: none;
}

small {
color: red;
}
</style>
<script src="https://cdnjs.cloudflare.com/ajax/libs/validator/13.9.0/validator.min.js"></script>
</head>
<body>

<h2>Signup Form</h2>

<form id="form">
<label>Email:</label><br>
<input type="text" id="email"><br>
<small id="emailError"></small><br><br>

<label>Phone Number:</label><br>
<input type="text" id="phone"><br>
<small id="phoneError"></small><br><br>

<button type="submit">Submit</button>
</form>
<script>
document.getElementById("form").addEventListener("submit", function(e) {
e.preventDefault();

let email = document.getElementById("email").value;
let phone = document.getElementById("phone").value;

let emailError = document.getElementById("emailError");
let phoneError = document.getElementById("phoneError");

emailError.innerText = "";
phoneError.innerText = "";

let valid = true;

if (!validator.isEmail(email)) {
emailError.innerText = "Invalid email address";
valid = false;
}
```

```
if (!validator.isMobilePhone(phone, 'any')) {
phoneError.innerText = "Invalid phone number";
valid = false;
}

if (valid) {
alert("Form submitted successfully!");
}
}});
</script>
</body>
</html>
```

Output Screen:



Explanation

- Validator.js is used for:
  - Email validation → isEmail()
  - Phone validation → isMobilePhone()
- When user enters invalid data:
  - Error message appears below input
- If valid:
  - Success alert is shown
- Prevents form submission using e.preventDefault()

RESULT

The input form was successfully designed with validation for email and phone number using HTML, CSS, JavaScript, and Validator.js. Appropriate error messages are displayed for invalid inputs.

CONCLUSION

Form validation improves data accuracy and user experience. Using Validator.js simplifies validation logic and ensures reliable input checking.